Model Evaluation Metrics

Standard formulas for R^2, RMSE and MAE

1. Coefficient of Determination (R^2)

$$R^2 = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}} = 1 - \frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{n} (y_i - \bar{y})^2}$$

R^2 is the proportion of the variance in the actual values that the model explains. $R^2 = 1$ is a perfect fit; $R^2 = 0$ means the model does no better than predicting the mean.

2. Root Mean Square Error (RMSE)

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2}$$

RMSE is the square root of the average squared prediction error, expressed in the same units as the data. Because errors are squared, large errors weigh more heavily. Lower is better.

3. Mean Absolute Error (MAE)

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^{n} \left| y_i - \hat{y}_i \right|$$

MAE is the average absolute difference between actual and predicted values, in the same units as the data. It is less sensitive to outliers than RMSE. Lower is better.

Where

y_i	Actual (observed) value of observation i
$\hat{y}_i$	Predicted value of observation i
$\bar{y} = \frac{1}{n} \sum_{i=1}^{n} y_i$	Mean of the actual values
n	Number of observations
SS_{res}	Residual sum of squares
SS_{tot}	Total sum of squares